Selection of k-th smallest element

(ADA-2023, CSE222)

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Overall algorithm

```
Select(A, n, k)
if n = 1 then
  |  Return A[1];
end
p ← ChoosePivot(A, n);
A_left ← {A[i] : A[i] < p};
A_right ← {A[i] : A[i] > p};
if |A_left| = k − 1 then
  |  Return p;
end
if |A_left| > k − 1 then
  |  Return Select(A_left, |A_left|, k);
end
else
  |  Return Select(A_right, |A_right|, k − |A_left| − 1);
end
```

k

$\boxed{O(n)}$

Recursive call in a smaller subproblem

$\frac{7n}{10}$

$\frac{7n}{10}$

```
ChoosePivot(A, n)
Split A into groups of
size g = ⌈n/5⌉ each
B_1, ..., B_⌈n/5⌉;
for i = 1, ..., g do
  |  Sort B_i using
  |  Merge Sort;
end
C ← {b_i : median of
B_i, i = 1, ..., g};
p ← Select(C, g, g/2);
Return p;
```

$\frac{n}{5}$

median of medians

- What are the subproblems? One is median of medians selection. Subproblem size is $\frac{n}{5}$.
  The other subproblem is in A_{left} (or in A_{right}).
- We have that $|A_{\mathsf{left}}| \leq \frac{7n}{10}$ and $|A_{\mathsf{right}}| \leq \frac{7n}{10}$.
- Recurrence is $T(n) = T(n/5) + T(7n/10) + cn$. $\boxed{T(1) = 1}$

Guess that $T(n) \leq dn$ w.l.g. assume that $d \geq 1$.

Base Case: $n = 1$. $T(1) = 1 \leq d$

Inductive Hypothesis: The guess is correct for all values $n = 1, 2, \ldots, k-1$.

(strong mathematical induction)

Induction Step: $n=k$.

$$T(k) = T\left(\frac{k}{5}\right) + T\left(\frac{7k}{10}\right) + ck$$

By induction hypothesis $T(k/5) \le \frac{dk}{5}$.

$$T\left(\frac{7k}{10}\right) \le \frac{7dk}{10}.$$

$$T(k) = T\left(\frac{k}{5}\right) + T\left(\frac{7k}{10}\right) + ck$$

$$\le \frac{dk}{5} + \frac{7dk}{10} + ck$$

$\frac{9dk}{10} + ck \le dk$

$ck \le \frac{dk}{10}$

Substitution Method

$$= \frac{9dk}{10} + ck \le dk$$

~~Hence~~ for all $d \ge 10c$

Substitution method

Reading material

- Jeff Erickson Lecture Notes Section 1.8.

Quiz Syllabus: 6th Feb in lecture time.

Duration: will announce. (MCQ Format).

Asymptotic notations Big-Oh, Big-Omega

Recurrences.

Divide and Conquer